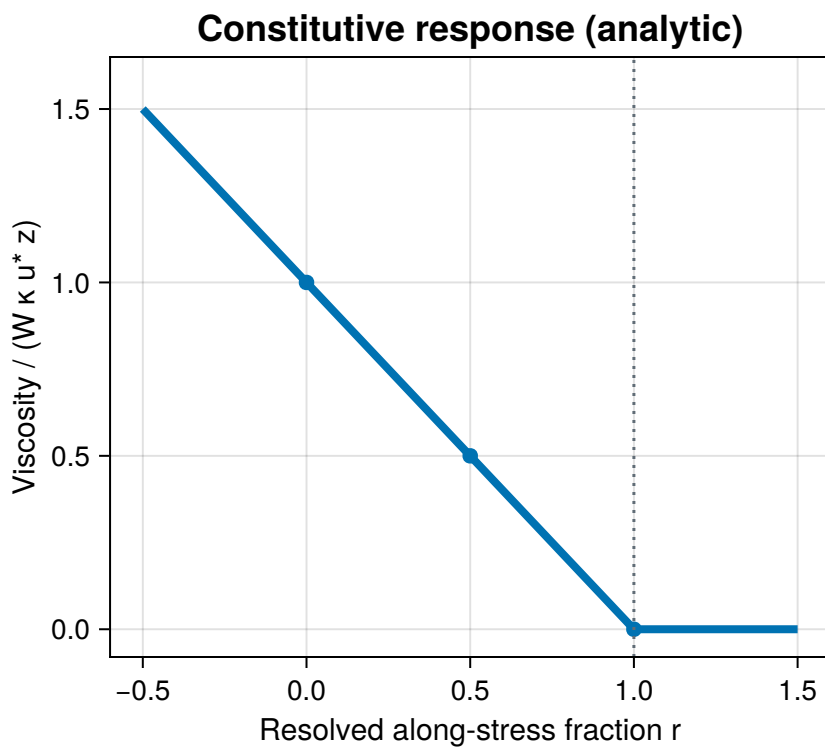


Can a thin surface closure improve coarse LES?

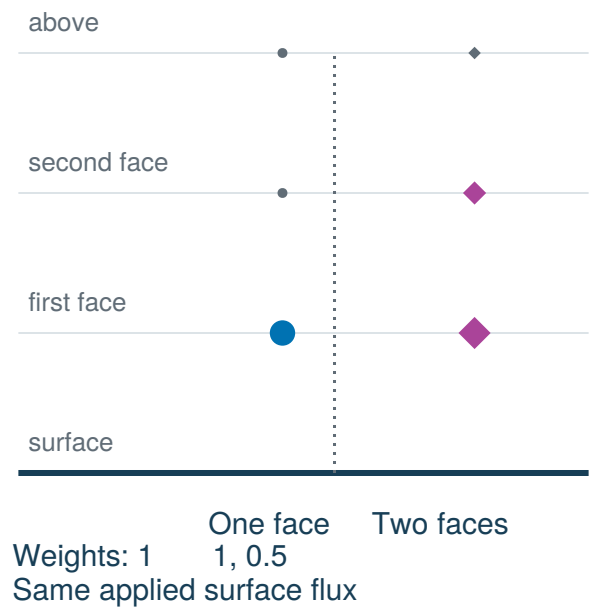
Supply unresolved near-wall transport while preserving resolved turbulence.
Four coarse GABLS1 comparisons admitted; transfer tests ongoing.

A local estimate of missing transport

Filter velocity, scalar, and wall-flux samples in physical time. Use their centered covariance to estimate the resolved share of the surface flux.



Shallow vertical support



What this formula does

No resolved stress: full neutral similarity viscosity. Half resolved: half the viscosity. Fully resolved or more: the correction turns off.
Opposite-sign transport can give a factor greater than one.

What it does not establish

The added flux still depends on the instantaneous vertical gradient.
The formula does not force exact flux balance, include stability functions, or measure the numerical transport produced by WENO.

One-face height: 12.5 m in both planned GABLS comparisons.
Scalar correction uses signed flux ratios; zero/tiny wall flux disables it.